
Course Instructor: **Nishat Tasnim**
Course: **CSE103 (Structured Programming)**

Lab: **09**

Topics Today

1. Function

Problem 1

Write a C program to input any number from user and find cube of the given number using function.

Problem 2

Write a C program to input two or more numbers from user and find maximum and minimum of the given numbers using functions.

Problem 3

Develop a function that takes two integers as parameters and returns the count of numbers that are divisible by 5 and 7 in that range. Call the function from main and print the result.

Problem 4

Write a function that takes the radius of a circle and returns the area of that circle. Write another function that takes area as parameter and determines whether the area is greater than 25 or not. If the area is greater than 25 then print "Greater than 25" or else "Less than 25". Call both functions from the main.

Problem 5

Write a function `integerPower(base, exponent)` that returns the value of $base^{exponent}$. For example, `integerPower( 3, 4 ) = 3 * 3 * 3 * 3`. Assume that exponent is a positive, nonzero integer,

and base is an integer. Function `integerPower` should be used to control the calculation. Do not use any math library functions.

Problem 6

The greatest common divisor (GCD) of two integers is the largest integer that evenly divides each of the two numbers. Write a function `gcd` that returns the greatest common divisor of two integers.

Problem 7

Take an integer number `N` from the user and calculate the summation of all natural numbers from 1 to `N`. You must use a **recursive** function to solve this problem. Example: `Add (5)` returns $1+2+3+4+5 = 15$, `Add (3)` returns $1+2+3 = 6$

Problem 8

Write a function that takes an integer `N` as parameter. And print Hello World `N` times using **recursion**. Remember that, you cannot use any loop in this program.